# ST. JOSEPH'S COLLEGE OF ENGINEERING

*(An Autonomous Institution) • OMR, Chennai - 119*

LAB RECORD

## Professional English Lab Integrated

**CLASS CODE: EN25C09**

### STUDENT DETAILS

**Student Name:** Jaish Sankar

**Roll No:** 26CS129

**Date:**

**Subject Name:** Professional English Lab Integrated

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| **Revision note** | This edition merges the lab-record cover with the personal cover, combines overlapping profile, SWOC, inspiration and NPTEL pages, and removes repeated worksheet text. Every distinct answer, definition and reflection from the original is kept. No personal English test scores are shown, because the scores in the original file belonged to another student. |

STUDENT PROFILE

## Profile & Introduction

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|---|---|
| **Student** | Jaish Sankar · Registration No. 26CS129 · First-year B.E. Computer Science & Engineering, St. Joseph's College of Engineering, Chennai (Batch 2026–2030) |
| **Technical direction** | AI, robotics, computer vision, embedded systems |
| **Ambition** | Build intelligent machines and, one day, a robotics company |
| **Present focus** | Programming fundamentals, mathematical reasoning and practical project work |
| **Technical skills in development** | **Programming:** Python, C and C++ · **Problem solving:** algorithmic thinking and introductory DSA · **Tools:** Linux and Git · **Supporting subjects:** mathematics, electronics, AI and computer vision |
| **Career interests** | Robotics, autonomous systems, intelligent automation, machine perception and the development of dependable engineering solutions. |
| **Current status** | I am still building my foundations. Future projects, internships and certifications are goals rather than completed credentials. |

### Who I am

I am Jaish Sankar, a first-year Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai. My main interest is how software and electronics can work together to create intelligent machines. I am concentrating on problem solving, programming in Python and C/C++, and learning the mathematics that supports computing. I also want to develop confidence in Linux, Git, data structures and algorithms. I hope to connect theoretical lessons with small experiments, carefully documented programs and practical design challenges.

### Personal pitch: introducing myself

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| **Opening** | Good morning. I am Jaish Sankar, currently pursuing my first year of B.E. Computer Science and Engineering at St. Joseph's College of Engineering, Chennai. |
| **Interests** | I am drawn to artificial intelligence, robotics, computer vision and embedded devices. I enjoy thinking about how a machine senses its environment, processes information and decides what to do next. |
| **Ambition** | My goal is to become an AI and Robotics Systems Engineer. In the longer term, I would like to establish a company that creates useful robotic technologies. To move toward that goal, I plan to learn steadily, complete meaningful projects and develop strong teamwork skills. |
| **Closing** | Thank you for the opportunity to introduce myself. |

## Career vision: engineering intelligent systems

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| **Professional direction** | I want to build systems that connect perception, reasoning and physical action. This means understanding software, sensors, controllers, machine learning and testing—not treating AI as a black box. |
| **Short-term priorities** | Improve confidence in Python and C/C++, practise problem solving, begin electronics fundamentals and document small programs. |
| **Long-term goals** | Move into machine learning, computer vision, ROS 2 and embedded robotics. Gain experience through competitions and internships before attempting advanced autonomous systems. |
| **Values** | Reliability, safety, careful experimentation, honest reporting and collaboration are as important as ambitious technical goals. |

SELF ASSESSMENT

# SWOC Self-Assessment

**Strengths** – Clear career direction; preference for understanding ideas deeply; interest in practical experiments; motivation to build and test projects.

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| **Purposeful learning** | A specific destination—AI and robotics engineering—helps me make thoughtful choices about what to study and practice. |
| **Systems thinking** | I prefer understanding the link between components rather than only following tutorials. Robotics rewards this habit because software and hardware must interact reliably. |
| **Initiative and curiosity** | My proposed Python voice assistant demonstrates the kind of practical challenge I want to explore: take a broad idea, divide it into modules and test each module. |
| **Critical perspective** | Inspiration should encourage ambition without replacing evidence. I want to examine engineering claims carefully and learn from limitations. |

**Weaknesses & improvement** – Early-stage knowledge in programming and electronics; the temptation to pursue too many topics simultaneously; little direct industrial or robot-building experience.

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| **Programming gap** | Action: solve small C/C++ and Python problems each week, review errors and maintain a log of concepts that need more practice. |
| **Too many interests** | Action: focus on one learning milestone at a time and complete a small demonstration before moving to the next subject. |
| **Limited hardware access** | Action: begin with affordable sensors or simulations, then plan a first physical build only after understanding its design. |
| **Professional exposure** | Action: develop clear project documentation, seek feedback from seniors and look for club, internship or research opportunities. |

**Opportunities** – Open-source development tools; NPTEL and other learning resources; engineering clubs, hackathons, competitions and research opportunities.

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| **Accessible learning** | Use structured study material, open-source software and practical demonstrations to study AI and robotics at a reasonable cost. |
| **Engineering communities** | Participate in collaborative projects, coding events or robotics competitions to learn teamwork and design communication. |

**Challenges** – Hardware expense; changing AI and robotics tools; competitive opportunities; balancing semester work with independent projects.

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| **Resource limitations** | Reuse parts, simulate first and evaluate every purchase in terms of learning value. |
| **Academic balance** | Maintain a realistic study routine so that personal exploration strengthens rather than interrupts course requirements. |

CAREER ROADMAP

# Career Roadmap & Inspiration

## Four-year learning plan

| Period | Stage | Focus | Target |
|---|---|---|---|
| **2026–27** | Foundation | Improve Python, C/C++, algorithmic reasoning, Linux, Git, mathematics and elementary electronics. | A well-documented software project and introductory robotics experiment. |
| **2027–28** | Intelligent perception | Explore machine learning, deep learning and computer vision with tools such as OpenCV and PyTorch. | An object-recognition or tracking demonstration. |
| **2028–29** | Robotics integration | Study ROS 2, sensor interfaces, navigation, state estimation and simulation. | A robot or simulated agent that can navigate and report its decisions. |
| **2029–30** | Advanced systems | Investigate reinforcement learning, edge AI and advanced embedded systems. | An integrated capstone project and a clearly presented engineering portfolio. |

## How the four stages connect

Each stage prepares the next. The foundation year (Python, C/C++, Linux, Git, mathematics and elementary electronics) is what makes machine learning and computer vision workable in year two. Perception skills then feed the navigation, sensor-interface and state-estimation work of ROS 2 in year three, and all of these come together in the capstone project and portfolio of the final year. Because each target is a demonstration that can be shown and tested, progress can be checked every year instead of being judged only at the end.

## Inspiration: Elon Musk and engineering ambition

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| **Why this example interests me** | Elon Musk's involvement in engineering-intensive businesses has encouraged me to think about challenging problems that join software, hardware and manufacturing. |
| **Lesson I draw** | Large goals become more realistic when complex systems are broken into testable technical problems. I want to practise first-principles reasoning rather than accept an idea without understanding it. |
| **Balanced reflection** | Ambitious timelines and public claims also need scrutiny. I want to combine bold ideas with transparent evidence, teamwork, safety and honest acknowledgment of uncertainty. |

## Inventors and responsible innovation

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| **The engineer I admire** | An engineer who investigates a problem patiently, tests a prototype thoroughly, and is willing to revise a design when the evidence changes. |
| **Working with others** | A useful robot rarely comes from one discipline. Good results require programmers, electronics designers, mechanical specialists and people who understand user needs. |
| **A principle for my work** | Create technology that serves a clear purpose, respects safety and privacy, and is understandable enough to maintain and improve. |

## My learning from role models

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|---|---|
| **Think beyond a subject** | Intelligent robotics combines computing, mathematics, communication and physical engineering. |
| **Stay practical** | A prototype with reliable performance is more valuable than an ambitious description without testing. |
| **Lead responsibly** | A future robotics founder must be able to listen, explain technical decisions and accept criticism. |
| **Focus on growth** | My present responsibility is not to appear experienced; it is to become competent through regular, measurable work. |

# NPTEL Study Notes (Weeks 1-8)

Course: Enhancing Soft Skills and Personality (IIT Kanpur, Prof. T. Ravichandran). These are expanded study notes derived from the lecture themes, with paraphrased personal diary reflections. They are not evidence of a certificate or course completion. Diary topic is the title used in my earlier personal diary for that week.

## WEEK 1 · LECTURE 1 Introduction 1: Highlights of Developing Soft Skills

| | |
|---|---|
| **Concept** | Soft skills influence how knowledge becomes useful in daily life. Communication, listening, responsibility and the ability to work with other people affect both study and employment. The introductory session connects these habits with individual character and the decision to improve deliberately. |
| **Key takeaways** | • Soft skills decide how well technical knowledge is explained and applied.<br>• Listening and responding honestly are professional skills, not fixed personality traits.<br>• Improvement starts with a deliberate decision and with observing my own habits. |
| **Why it matters** | Technical ability is not enough on its own. An accurate explanation of a solution, an honest response to criticism, and a calm conversation in a team are also evidence of professional growth. |
| **My understanding** | This discussion made me compare the way I think about a challenge with the steps I actually take to solve it. |
| **Link to my studies** | I want to work toward AI and robotics systems, so I need to explain technical decisions in understandable language. For example, when presenting a small sensor-controlled robot or a C programming exercise, I should describe the problem, my method and what I learned. |
| **Worked example** | In a group lab session, instead of defending my wiring idea at once, I restate my partner's concern (“you think the sensor may be reading noise?”), explain my reasoning, and we agree on one test to settle it. |
| **Practical action** | Notice how I respond when someone disagrees with an idea; practise listening first and summarising their point before answering. |
| **Self-check questions** | 1. What are the parts of the communication loop?<br>2. When did I last respond defensively to feedback, and what would a better reply have been?<br>3. Can I explain my current project in three sentences to a non-technical listener? |
| **This week's checklist** | ☐ Summarise one disagreement before replying   ☐ Write a three-sentence project explanation   ☐ Ask a classmate for feedback on it |
| **Diary note** | Communication and Perception — Communication involves a sender, a message, a channel, a listener and feedback. Assumptions and different perspectives can change how an explanation is understood. *Application:* When presenting a robotics idea, I will ask the audience to restate the main goal or invite questions. |

## WEEK 2 · LECTURE 2 Introduction 2: Further Highlights of Developing Soft Skills

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| **Concept** | The second introductory lecture builds on personal development. It places attention on self-awareness, preparation, ethical conduct, listening, communication and the willingness to examine one's own behaviour. Progress is more realistic when a person observes habits instead of relying only on good intentions. |
| **Key takeaways** | • Self-awareness and preparation come before effective communication.<br>• Every conversation needs a purpose, an audience and an expected result.<br>• Checking understanding prevents wasted effort in teamwork. |
| **Why it matters** | A conversation has a purpose, an audience and an expected result. Choosing precise words and checking understanding makes both presentations and everyday teamwork more effective. |
| **My understanding** | I can relate this topic to preparing for a technical career while still developing my fundamentals. |
| **Link to my studies** | As I strengthen programming, mathematics and robotics, I can practise concise presentations and active listening. A good team discussion should leave everyone knowing the next action, not merely the words that were spoken. |
| **Worked example** | Before a team meeting on the voice assistant, I write the purpose (agree the command list), the audience (my teammates) and the result (five commands with an owner each), and I close by repeating the next actions. |
| **Practical action** | After one class discussion, write down what I explained clearly, what confused others, and one small adjustment for next time. |
| **Self-check questions** | 1. What was the purpose of my last discussion, and did it end with a next action?<br>2. Which words did I use that others may not have understood?<br>3. What one adjustment will I try next time? |
| **This week's checklist** | ☐ Write purpose, audience and result for the next meeting   ☐ Log one clear and one confusing explanation   ☐ End a discussion by stating next actions |
| **Diary note** | Presentation and Visual Communication — A useful presentation has a clear introduction, organised points and a conclusion. Diagrams should support the message instead of crowding slides with text. *Application:* I will use a block diagram and one live demonstration to explain the voice-assistant proposal. |

## WEEK 3 · LECTURE 3 Mindset 1: Definition and Types

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| **Concept** | A mindset refers to the beliefs and assumptions through which a person interprets experiences. A positive outlook encourages effort, but an unexamined negative assumption can discourage action before a problem is even attempted. Mindsets influence reactions to success, mistakes, advice and unfamiliar situations. |
| **Key takeaways** | • A mindset is a set of beliefs that shapes reactions to success, mistakes and advice.<br>• A limiting belief can stop action before any attempt is made.<br>• Acknowledge the difficulty, then choose a smaller, achievable step. |
| **Why it matters** | Recognising a limiting belief is different from pretending difficulties do not exist. I can acknowledge that a topic is challenging and still choose to attempt a smaller, achievable step. |
| **My understanding** | The lecture showed me that knowing a rule and consistently following it are two different things. |
| **Link to my studies** | When progress on a small sensor-controlled robot or a C programming exercise is slow, I should question the assumption that a first attempt must be perfect. Breaking a complex problem into smaller parts makes the challenge easier to investigate. |
| **Worked example** | “I am bad at C pointers” becomes “I have not yet practised pointers; this week I will trace three short programs by hand and then test them.” |
| **Practical action** | Identify one sentence I use when a task feels hard, then rewrite it as a realistic statement about a skill I can practise. |
| **Self-check questions** | 1. Which belief slows me down most?<br>2. How can I rewrite it as a realistic statement?<br>3. What is the smallest step I can take today? |
| **This week's checklist** | ☐ Write down one limiting sentence   ☐ Rewrite it realistically   ☐ Complete one small step on a hard topic |
| **Diary note** | Active Listening and Body Language — Listening requires attention and confirmation, not simply hearing words. Posture, gestures and facial expression also influence professional communication. *Application:* During a team meeting I will summarise the requirement before proposing a technical solution. |

| WEEK 4 · LECTURE 4 Mindset 2: Learning Mindsets | |
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| **Concept** | The learning-mindset lecture distinguishes responses that close off improvement from responses that welcome it. A fixed outlook may treat errors as proof of inability; a growth-oriented approach treats them as feedback about a method, strategy or level of preparation. |
| **Key takeaways** | • A fixed outlook treats errors as proof of inability; a growth outlook treats them as feedback.<br>• Improvement needs several attempts, good feedback and correction.<br>• The evidence of progress is a better method after each attempt. |
| **Why it matters** | Progress in a difficult subject usually requires several attempts, good feedback and corrections. This does not guarantee immediate success, but it encourages sustained, thoughtful practice. |
| **My understanding** | I would describe the principle through the choices I make each day, rather than through a single examination result. |
| **Link to my studies** | If I cannot solve a task using programming, mathematics and robotics immediately, I will experiment with examples, read feedback carefully and retry. The main evidence of improvement is a better method after each attempt. |
| **Worked example** | A loop program prints the wrong output. I record the cause (an off-by-one error in the condition), try a second method (tracing the values in a table) and note how my approach changed. |
| **Practical action** | When an exercise fails, record the cause, try a second method, and describe what changed rather than merely recording a wrong answer. |
| **Self-check questions** | 1. What caused my last failed exercise?<br>2. What second method could I have tried?<br>3. How can I show improvement beyond a score? |
| **This week's checklist** | ☐ Keep an error log for one week   ☐ Try a second method before asking for the answer   ☐ Review the log on Sunday |
| **Diary note** | Teamwork and Communication Barriers — Different assumptions, noise and unclear roles create misunderstandings. Cooperative work improves when responsibilities and expectations are discussed early. *Application:* For a robotics team task I will clarify who handles software, hardware integration and testing. |

## WEEK 5 · LECTURE 5 Mindset 3: Secrets of Developing Growth Mindsets

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| **Concept** | This session focuses on building a growth mindset through purposeful effort, reflection and practical goals. Improvement becomes more visible when long ambitions are divided into small tasks and when feedback is used to modify the approach rather than judge personal worth. |
| **Key takeaways** | • A growth mindset is built through purposeful effort, reflection and practical goals.<br>• Long ambitions become visible progress when split into small tasks.<br>• The word “yet” helps only when backed by repeated practice and honest review. |
| **Why it matters** | The word “yet” helps reframe an unfinished skill, but a useful growth mindset also requires evidence: repeated practice, specific study methods and honest review of results. |
| **My understanding** | The material encouraged me to work deliberately, review my behaviour and make one measurable adjustment at a time. |
| **Link to my studies** | My plan to develop in AI and robotics systems benefits from frequent practice. I can track a practical target, such as solving a set of small coding problems or improving one feature in a small sensor-controlled robot or a C programming exercise. |
| **Worked example** | Weekly goal: solve three array problems. Attempt 1 takes 40 minutes with hints; attempt 3 takes 15 minutes without hints. The comparison between the two is my evidence of growth. |
| **Practical action** | Set one weekly skill goal, complete three practice attempts and compare the final attempt with the first. |
| **Self-check questions** | 1. What is my one skill goal this week?<br>2. How will I compare my first and final attempts?<br>3. What feedback will I use to change my method? |
| **This week's checklist** | ☐ Set one weekly skill goal   ☐ Make three practice attempts   ☐ Compare the last attempt with the first |
| **Diary note** | Intercultural Communication and Conflict — Good collaboration recognises differences in habits and communication styles. Disagreements can be resolved by identifying a shared problem and discussing evidence calmly. *Application:* I will avoid personal criticism and compare possible solutions against common project goals. |

## WEEK 6 · LECTURE 6 Managing Time 1: Importance and Perceptions of Time

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| **Concept** | Time is limited, and different activities compete for it. This lecture asks learners to consider how they perceive time and where their hours go. Without first observing real habits, a timetable may become unrealistic or fail to account for interruptions. |
| **Key takeaways** | • Time is limited and different activities compete for it.<br>• Observe real habits before building a timetable.<br>• A plan should balance urgent work, long-term learning, breaks and responsibilities, with room to adjust. |
| **Why it matters** | Time management starts with awareness. Planning can balance urgent work, long-term learning, breaks and responsibilities while leaving room to adjust when a task takes longer than expected. |
| **My understanding** | I understood the central message as a reminder to turn an idea into a habit, not simply memorise a definition. |
| **Link to my studies** | When juggling classes and programming, mathematics and robotics, I need an honest picture of my schedule. I can compare planned study time with what I actually use, then adjust without removing necessary rest. |
| **Worked example** | Recording a Saturday in half-hour blocks shows two hours lost to switching between tasks. I then reserve a fixed morning block for programming practice and keep the evening free for rest. |
| **Practical action** | Record one study day in half-hour blocks, identify avoidable delays and reserve a fixed period for a priority task. |
| **Self-check questions** | 1. Where did my hours really go yesterday?<br>2. Which delays were avoidable?<br>3. What fixed period will protect my priority task? |
| **This week's checklist** | ☐ Record one study day in half-hour blocks   ☐ Mark the avoidable delays   ☐ Reserve one fixed priority period |
| **Diary note** | Emotional Intelligence and Time Planning — Self-awareness, empathy and stress management support consistent study. Complex work becomes less intimidating when split into achievable steps. *Application:* I will divide programming practice into short sessions and review progress weekly. |

## WEEK 7 · LECTURE 7 Managing Time 2: Using Time Efficiently

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| **Concept** | Efficient time use means selecting priorities, estimating effort and limiting distractions. Working for many hours is not the same as producing good work. Concentrated sessions, realistic deadlines and checking completed tasks can improve the use of available time. |
| **Key takeaways** | • Efficiency means choosing priorities, estimating effort and limiting distractions.<br>• Many hours of work are not the same as good work.<br>• Every session should name a concrete output, such as a tested function. |
| **Why it matters** | A short task list works best when it names a concrete output rather than an unclear intention. Reviewing progress also reveals which tasks repeatedly need more time. |
| **My understanding** | The main lesson for me is that a useful skill appears in ordinary actions, especially when circumstances become demanding. |
| **Link to my studies** | To make steady progress toward AI and robotics systems, I should rank important tasks, avoid constant switching and set a visible result for each study session, such as a tested function or a written explanation. |
| **Worked example** | Tomorrow's three tasks: finish one C function (45 min), write its test (30 min), read one NPTEL note (20 min). Afterwards I compare the estimated and actual times to improve my next plan. |
| **Practical action** | Choose three tasks for tomorrow, estimate minutes for each, finish the most important one early and review actual times. |
| **Self-check questions** | 1. What concrete output will tonight's study session produce?<br>2. Which task repeatedly takes longer than I estimate?<br>3. Which distraction can I remove? |
| **This week's checklist** | ☐ Choose three tasks   ☐ Estimate minutes for each   ☐ Finish the most important one first   ☐ Review the actual times |
| **Diary note** | Critical Thinking, Leadership and Ethics — Strong decisions depend on examining assumptions and evidence. Responsible leaders listen, share credit and communicate limits honestly. *Application:* I will test project claims carefully and report accuracy, errors and restrictions rather than exaggerating performance. |

ENGLISH DEVELOPMENT

# English Development & Assessment

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| **Communication goals** | Explain a technical idea in a logical order, choose clear vocabulary, listen closely to questions and give concise answers. |
| **Relevant situations** | Introducing myself in a seminar, describing a project proposal, asking for laboratory support and preparing a short progress update. |
| **Practice approach** | Use sample conversations, sentence rewriting and vocabulary tasks. The examples below are learning exercises, not formal assessment results. |

## Interchange & conversation practice

| Situation | Question | My answer |
|---|---|---|
| **Introducing a project** | What are you proposing? | I plan to create a voice-controlled desktop assistant that responds to a small set of spoken commands. |
| **Discussing the past** | What did you study yesterday? | I revised C programming and practised conditional statements. |
| **Talking about plans** | What will you learn next? | I will strengthen my Python skills before moving to machine learning. |
| **Checking understanding** | Could you explain how it works? | The system receives an audio command, recognises the words, chooses an action and gives a spoken reply. |

## My communication reflection

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| **More precise explanations** | Describing a technical system in plain English encourages me to distinguish between the goal, the method and the expected outcome. |
| **Confidence in presentations** | Preparing a beginning, middle and closing for my personal introduction makes public speaking feel more manageable. |
| **Better listening** | When discussing projects, I want to confirm requirements before writing code or choosing hardware. |
| **Next improvement** | I will practise speaking for two minutes on a technical topic and review whether my explanation is accurate and easy to follow. |

## English assessment record

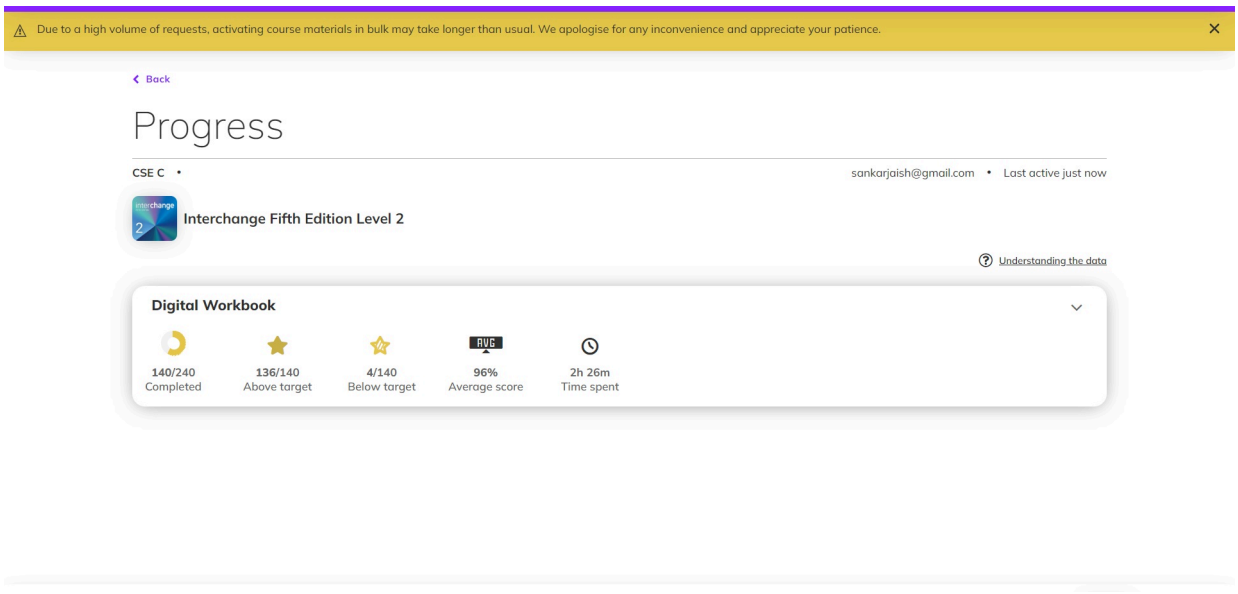
| | |
|---|---|
| **Personal test scores** | Not supplied in the uploaded portfolio. One page in the original file listed scores from another student as reference information; those do not show Jaish Sankar's proficiency and are not reproduced here. |
| **Development plan** | • Read a technical paragraph aloud, record it, then listen for pronunciation and pacing.<br>• Write a 150-word explanation of a small robotics or programming concept.<br>• Practise a one-minute introduction explaining a problem, approach and expected result.<br>• Add verified results from Preply, Speak & Improve or a reading test when personally completed. |

ANSWERED PRACTICE

# Personal Practice Worksheets (9)

Fill-in-the-blank answers are shown underlined. The key-word sentences are new examples that replace the identical placeholder sentence that appeared on every page of the original.

EN25C09 | Cambridge One Portal Assignment



# Professional English Practice Worksheets – Solved

*Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.*

## Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must
5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing
11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:** The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

## Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   **Answer (Screen Size):** A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".)

   (b) **Answer:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully

10. (a) **Answer:** can

    (b) **Answer:** must not

## Worksheet 1 – Part C (5 × 13 = 65 Marks)

### Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for every account.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.
5. Check that a website address begins with "https" and shows a padlock before entering any sensitive information.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or the National Cyber Crime Helpline (1930).

### Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

**Selected Question: 27.a**

**Bharatanatyam: The Dance That Carries Tamil Nadu's Soul**

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student's formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

### Question 28 (b) Formal letter of request to the Principal

**Selected Question: 28.b**

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal
St. Joseph's College of Engineering

OMR, Chennai – 600 119

**Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies**

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission.

Thank you for your kind consideration.

Yours obediently,
Inba Ilakiyan
(Register No.: [Register No.])

## Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

**Selected Question: 29.a**

**AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education**

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

## Question 30 (b) Personal email: Orientation programme

**Selected Question: 30.b**

**To:** [friend's email address]

**Subject:** My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Inba

## Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

### Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

### Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

### Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

### Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

### Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

### Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

### Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

### Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

### Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

### Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

### Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

### Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

### Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

### Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

### Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

## Worksheet 2 – Part B (Exercises 1 to 10)

### Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.

4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

### Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

### Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

### Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.

   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.

   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound.

   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.

   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message.

   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

### Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.

2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

### Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

### Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

### Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

### Exercise 9: Adverbs of Manner

1. (a) automatically
   (b) flawlessly
2. (a) accurately

(b) instantly

3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

### Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

## Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

### Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

### Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in working condition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns and accidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors before disconnecting the circuit.

### Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping the machine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the first cut.
8. Never leave the machine unattended or operate it without the instructor's permission.

## Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

### Question 1: A traditional harvest festival and eco-friendly practices

**Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit**

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

### Question 2: A temple car festival and sustainable measures

**The Temple Car Festival: Faith, Food and a Cleaner Celebration**

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

### Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

**Folk Art and Handicraft Fairs: Heritage with a Green Future**

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

## Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

### Question 1: Technical study tour to a major hydroelectric dam

**A Technical Study Tour to the Mettur Hydroelectric Dam**

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30 a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

### Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

**A Technical Study Tour to a Solar and Wind Hybrid Energy Farm**

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and

night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

### Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

**A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant**

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

## Worksheet 2 – Topic 4: Formal Letters of Request

### Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

**Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon**

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Inba Ilakiyan
(On behalf of the project team)

### Question 2: Letter to the Plant Manager (industrial plant visit)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

**Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students**

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Inba Ilakiyan

### Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

**Subject: Request for permission and machine shop access after college hours to build an electric go-kart**

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Inba Ilakiyan
(On behalf of the team)

### Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

**Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo**

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo, to be held on [dates] at [venue]. We respectfully request financial sponsorship and duty leave for this event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us valuable feedback from experts and will bring recognition to our college.

We request sponsorship for travel, accommodation, registration and prototype transport, which we estimate at ₹[amount], and duty leave for [number] days so that we do not lose attendance. We will make up for the missed classes and submit a report after the event.

I would be grateful for your kind support and approval.

Thank you.

Yours obediently,
Inba Ilakiyan
(On behalf of the team)

## Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

### Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

**Electric Vehicles versus Conventional Internal Combustion Engine Vehicles**

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV

produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

## Question 2: Remote/Virtual Internships versus On-Site Internships

**Remote Internships versus On-Site Internships for First-Year Engineering Students**

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

## Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

**Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises**

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws

carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

### Question 4: 3D Printing versus Traditional CNC Milling

**3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication**

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

## Worksheet 2 – Topic 6: Personal Email Writing

### Question 1: Email to a high school mentor/teacher about the first semester

**To:** [teacher's email address]

**Subject:** My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Inba Ilakiyan

### Question 2: Email to a high school science teacher about the first term

**To:** [science teacher's email address]

**Subject:** Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
Inba Ilakiyan

### Question 3: Email to an elder sibling/alumnus seeking advice

**To:** [sibling's or alumnus's email address]

**Subject:** Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that, I go to my technical club meetings for about two hours, and in the evening I try to study and finish my assignments. By the time I am free, I am usually very tired.

My biggest challenge is managing the lab record submissions. Because the records must be completed every week, they pile up, and I often finish them at the last minute, which affects the quality.

Since you have handled the same situation, could you please tell me how you balanced your academics with the clubs? I would also like your tips for semester exam revision, such as how to plan the timetable, which subjects to start with, and how to use previous question papers.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,
Inba

## Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

### Question 1: 4-day tour to an aerospace and satellite launch station

**A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station**

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

**Budget (per student):** train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

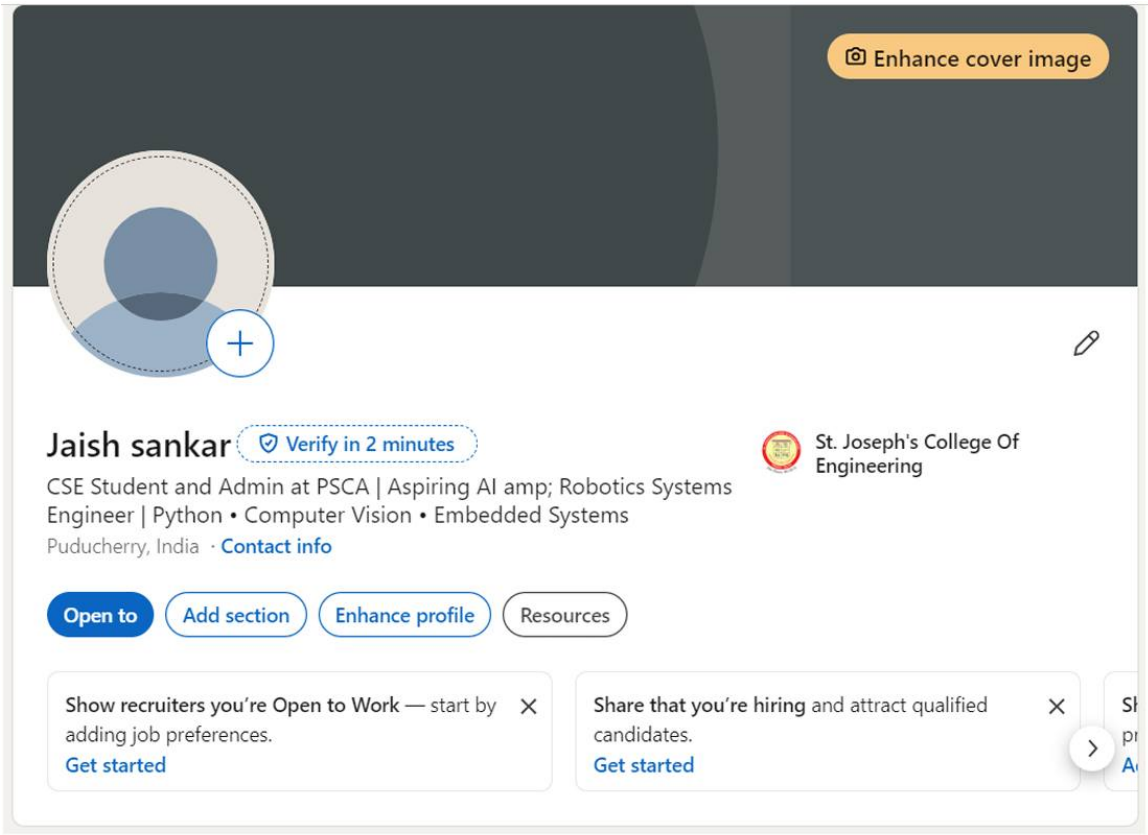
NPTEL LEARNING DIARY

# WEEK 8 | LECTURES 36-41: Ethics, Etiquette and Personal Development

Course: Enhancing Soft Skills and Personality (IIT Kanpur, Prof. T. Ravichandran). Week 8 covers ethics and etiquette, business etiquette, memory development, care for the environment and a review of the course.

| | |
|---|---|
| **Concept** | Ethics guides decisions about what is responsible and fair. Etiquette is the respectful behaviour expected in social and professional situations. Business etiquette applies these principles to workplace communication, meetings and relationships. |
| **Key takeaways** | • Ethical conduct requires honesty, responsibility and respect.<br>• Business etiquette includes punctuality, suitable communication and professional behaviour.<br>• Memory can be supported through attention, organisation, revision and meaningful associations.<br>• Care for the environment is part of responsible personal and professional conduct. |
| **Why it matters** | Technical knowledge is more useful when people can trust the person applying it. Respectful conduct supports teamwork, while good memory habits and environmental awareness improve everyday effectiveness. |
| **My understanding** | This week connects personal character with professional behaviour. A good engineer should not only solve a technical problem but also communicate honestly, respect others and consider the effect of a solution. |
| **Link to my studies** | In a CSE team project, I should acknowledge each member’s contribution, communicate politely, arrive prepared for meetings and protect shared information. I can remember technical concepts by making short notes and revising them regularly. I should also avoid unnecessary energy use and electronic waste. |
| **Worked example** | Before a project review, I arrive on time, prepare the latest test results, listen without interrupting and give accurate updates—including problems that remain unresolved. I keep a short revision sheet of key concepts and reuse or recycle electronic components responsibly. |
| **Practical action** | Follow one professional etiquette habit each day, revise one set of technical notes using spaced practice, and identify one practical way to reduce waste or energy use. |
| **Self-check questions** | 1. How are ethics and etiquette different?<br>2. What behaviours show good business etiquette?<br>3. Which revision method helps me remember technical concepts?<br>4. How can I reduce waste in my daily student life? |
| **This week's checklist** | ☐ Be punctual and prepared for a meeting<br>☐ Practise respectful listening and clear communication<br>☐ Review notes using spaced revision<br>☐ Take one action to reduce waste or energy use |
| **Diary note** | Course reflection — Soft skills develop through consistent choices. Ethical behaviour, professional etiquette, memory habits and environmental responsibility support both personal growth and effective teamwork. Application: I will combine technical learning with honest communication, respectful collaboration and responsible use of resources. |

# LINKEDIN PROFILE - Jaish Sankar




Name: Jaish Sankar

CSE Student and Admin at PSCA | Aspiring AI & Robotics Systems Engineer

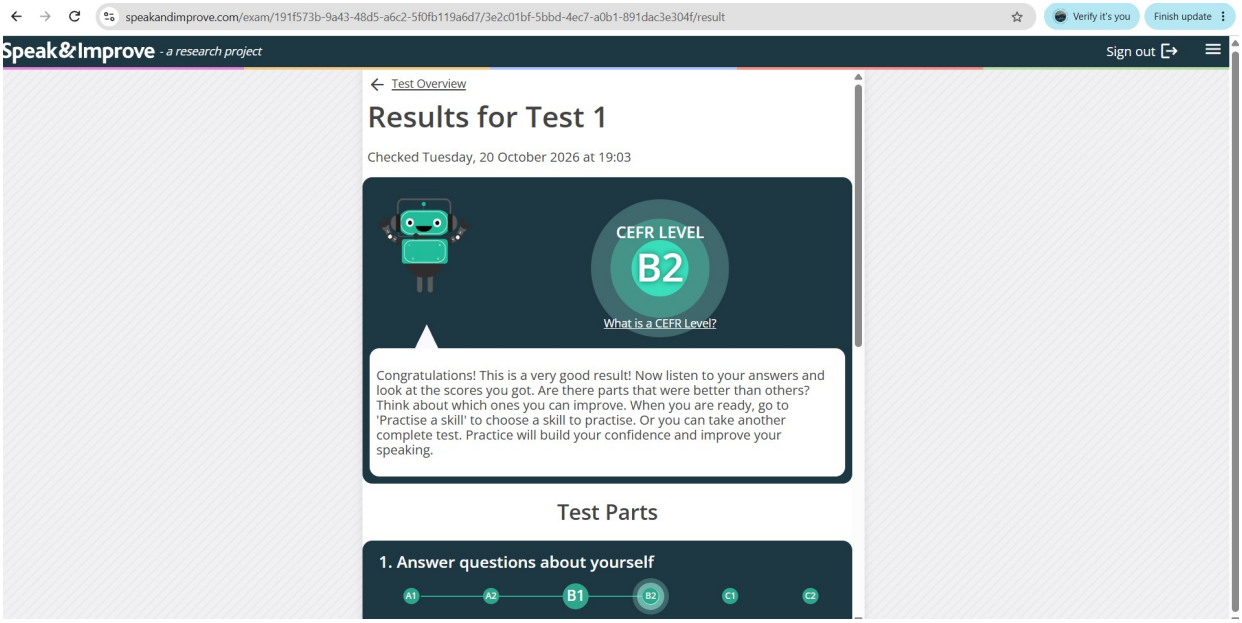
Skills: Python | Computer Vision | Embedded Systems

Institution: St. Joseph's College of Engineering

Location: Puducherry, India

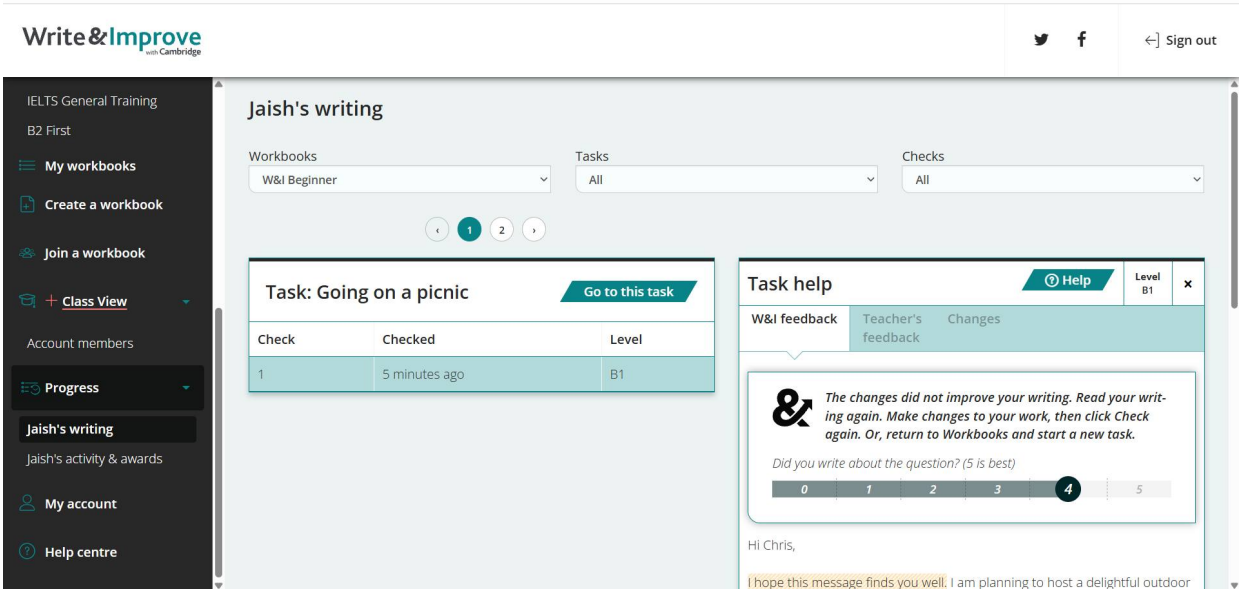
# READ & IMPROVE / SPEAK & IMPROVE

Cambridge English Assessment - Results Overview



# WRITE & IMPROVE

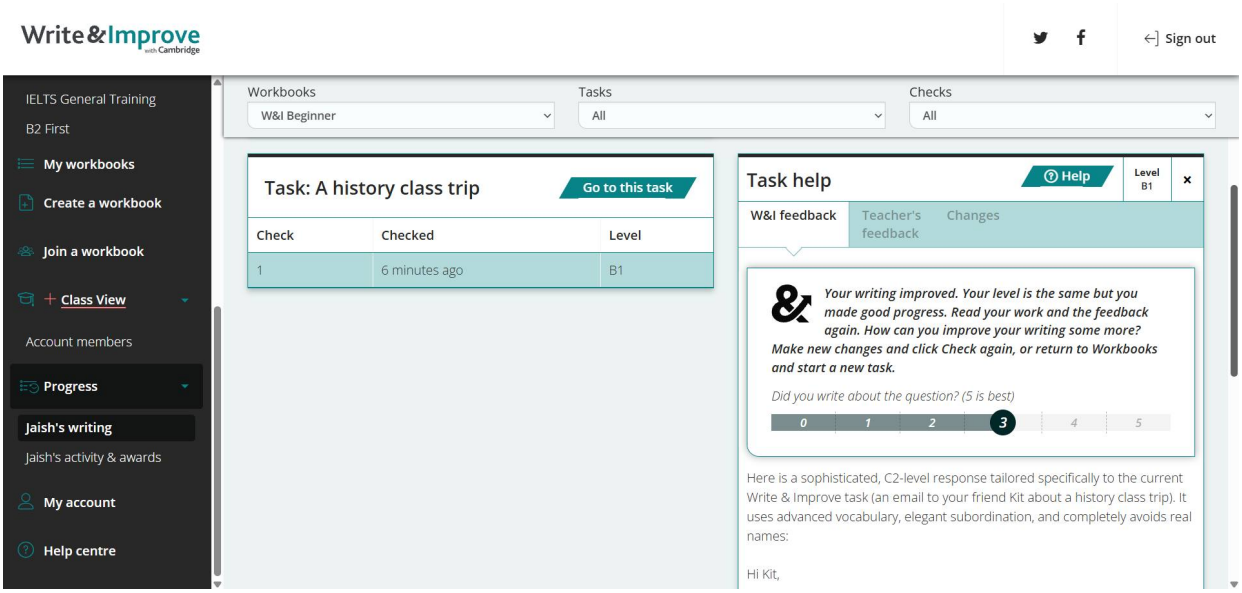
With Cambridge - Tasks Progress | B1 & A2 Levels




Task: Going on a picnic - Level B1 - Checked 5 mins ago

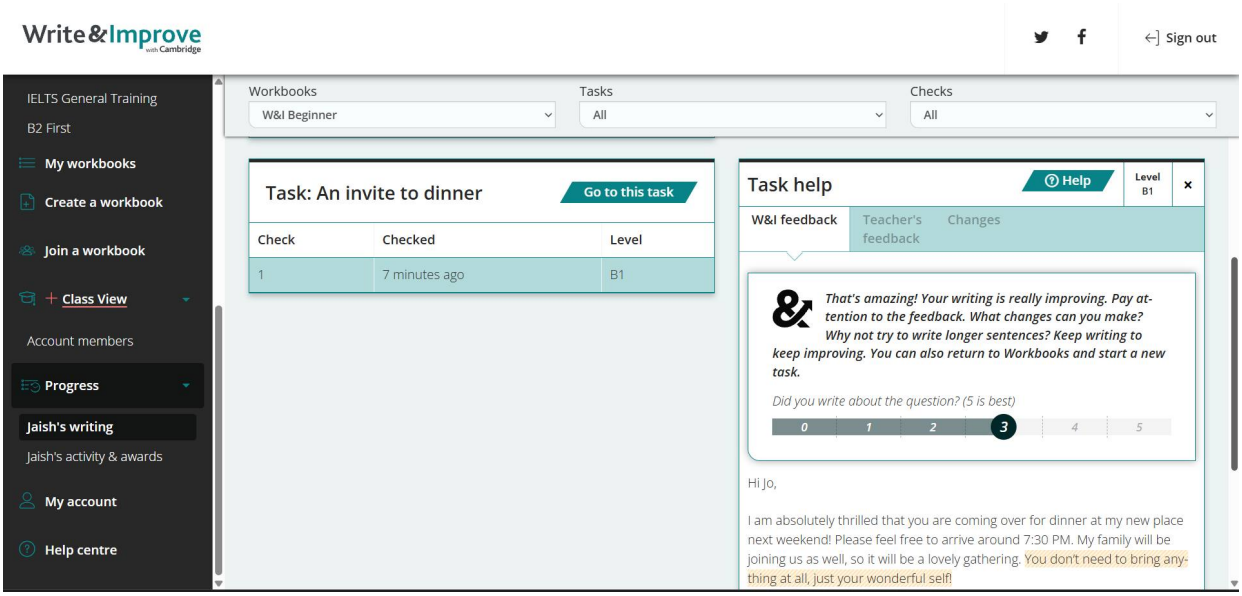
# WRITE & IMPROVE

Task 2 of 4



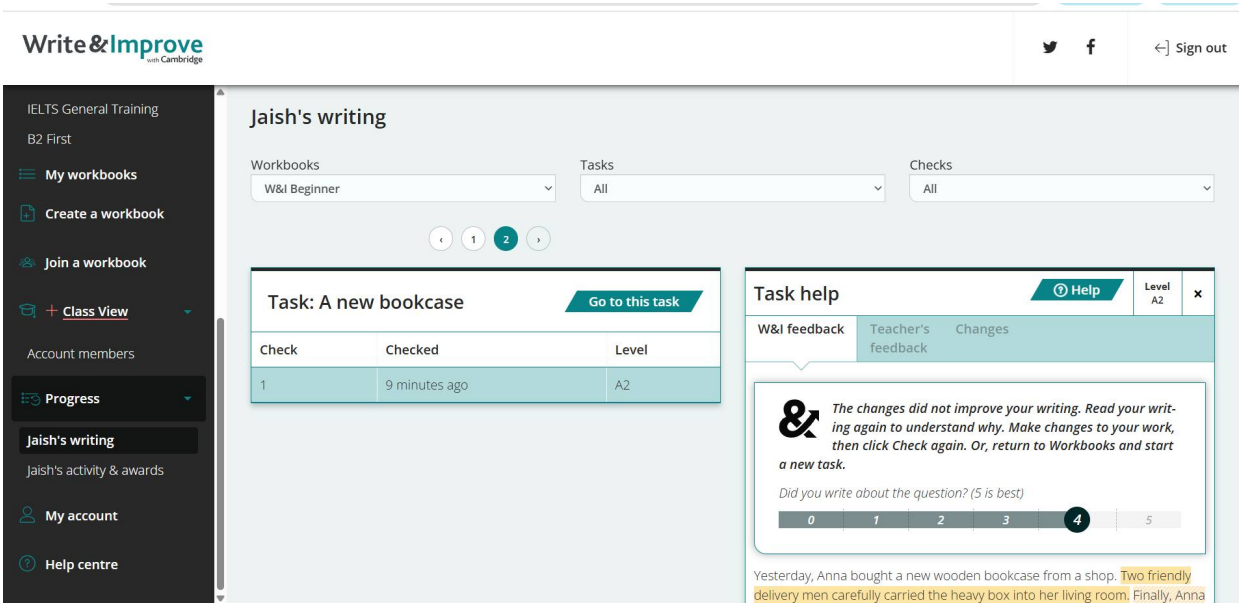
# WRITE & IMPROVE

Task 3 of 4



# WRITE & IMPROVE

Task 4 of 4



# Combined Progress Report

Read & Improve (Speak & Improve) - CEFR B2

Write & Improve - 4 Tasks Completed (B1, A2)

All tasks checked on 20 October 2026